

PREDIKSI STABILITAS RANTAI PASOK BERBASIS *TIME-SERIES* MENGGUNAKAN *LONG SHORT-TERM MEMORY (LSTM)*

Oleh:

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**S1 SAINS DATA
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LATAR BELAKANG

Stabilitas Rantai Pasok

tidak stabil menyebabkan kerugian

Rantai Pasok Rentan

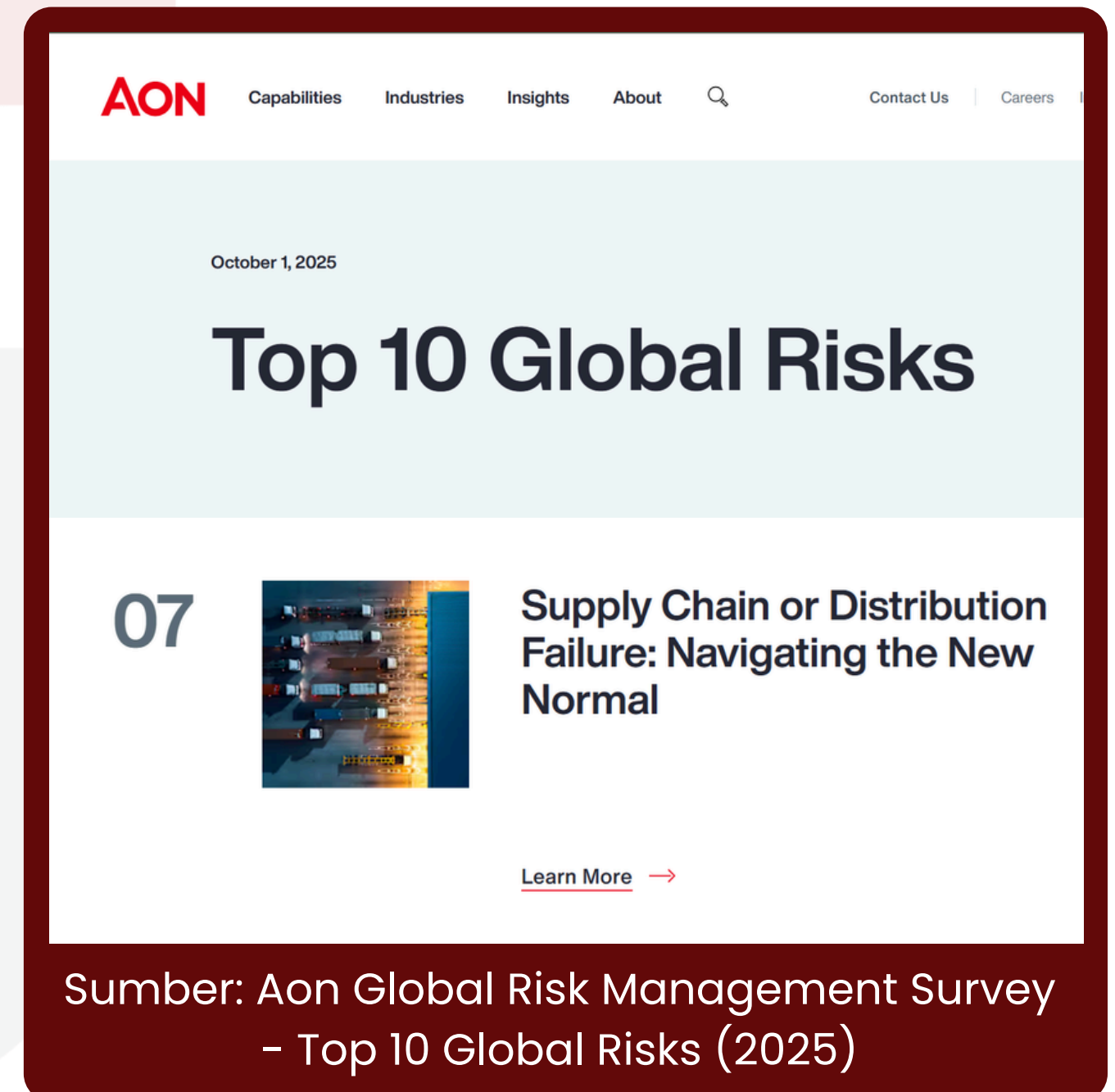
perusahaan butuh model prediksi risiko

Penelitian Sebelumnya

(Banerjee et al., 2020)

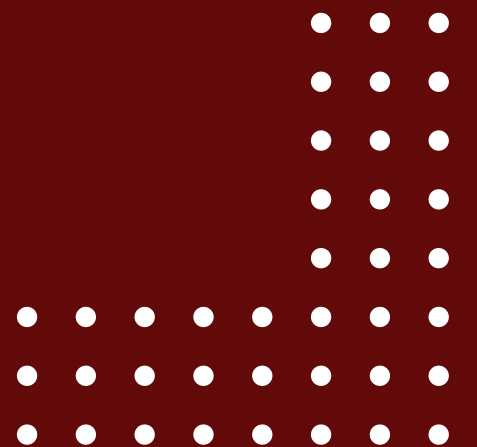
(Shreyas et al, 2025)

LSTM Prediksi *Multiclass*



RUMUSAN MASALAH

1. Bagaimana mempersiapkan data *time-series* rantai pasok untuk digunakan dalam pelatihan model LSTM?
2. Bagaimana performa model LSTM dalam memprediksi kategori stabilitas rantai pasok?



- 1 Stabilitas Rantai Pasok
- 2 Data Deret Waktu (*Time-Series* Data)
- 3 Pra-pemrosesan Data
- 4 Model LSTM

KONSEP / TEORI UTAMA

3 Pra-pemrosesan Data

- Interpolasi berbasis waktu [1]

$$y_t = y_{t_1} + (t - t_1) \left(\frac{y_{t_2} - y_{t_1}}{t_2 - t_1} \right)$$

- IQR dan Quantile Capping
- Normalisasi Min-Max [2]

$$x'_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

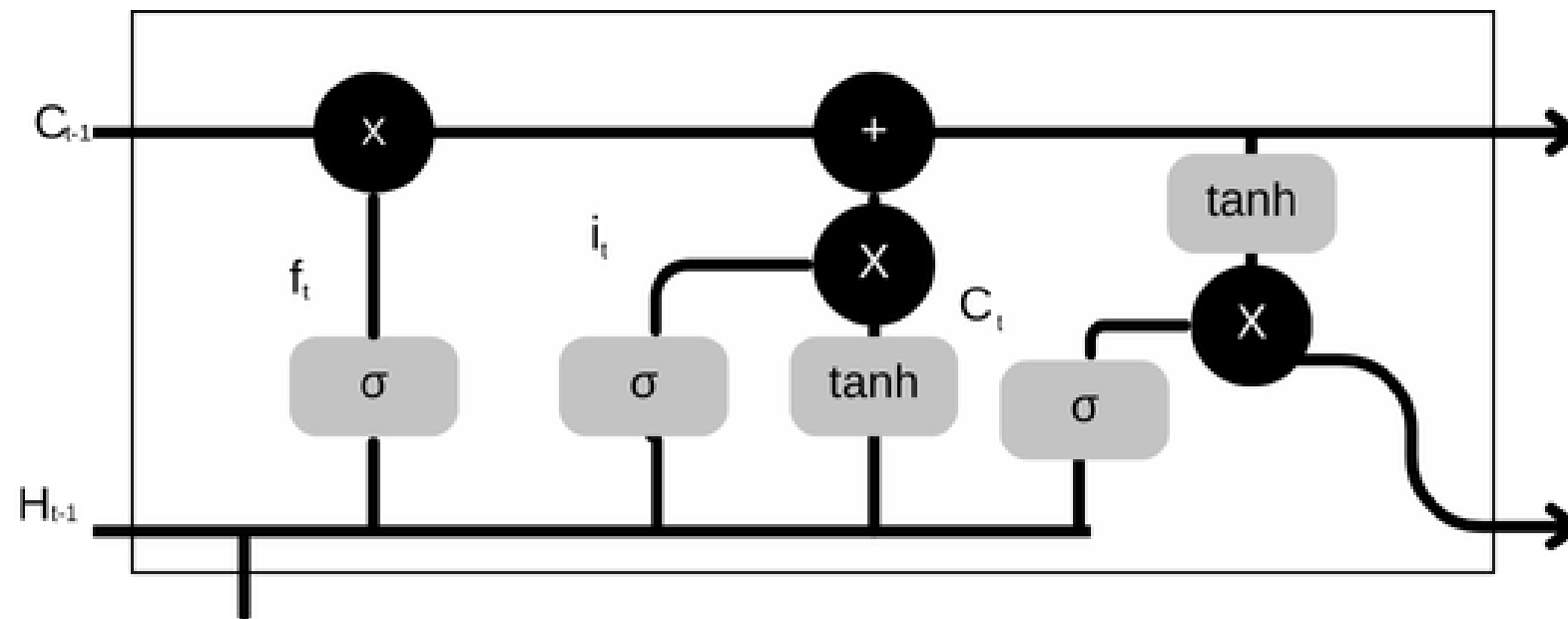
- Normalisasi Robust [2]

$$x'_i = \frac{x_i - Q_2(x)}{Q_3(x) - Q_1(x)}$$

- Balancing Augmentasi [3]
 - *Jittering*
 - *Drift*

4 Model LSTM

Arsitektur LSTM [1]



- Forget Gate [2]

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

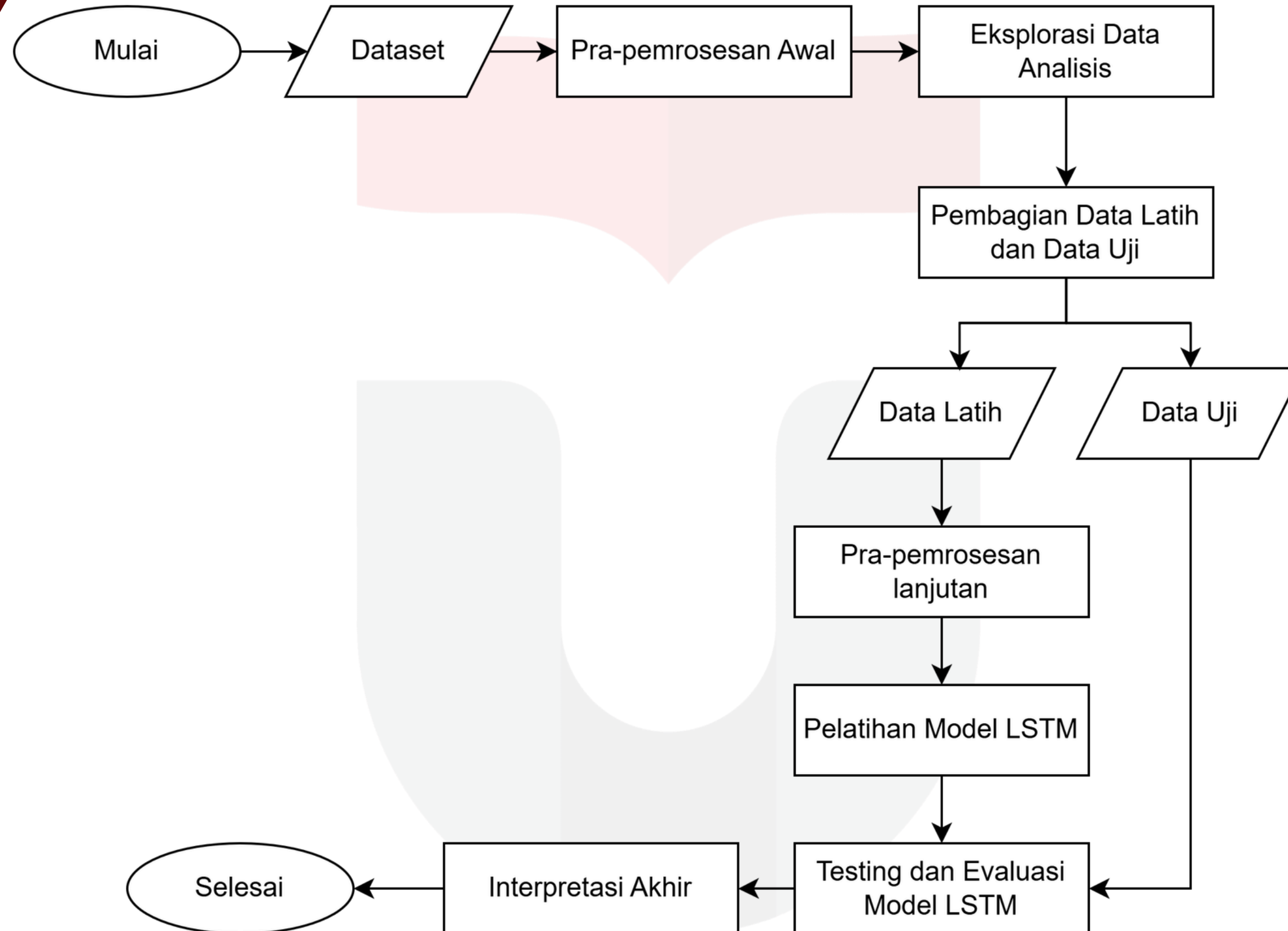
- Input Gate [2]

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

- Output Gate [2]

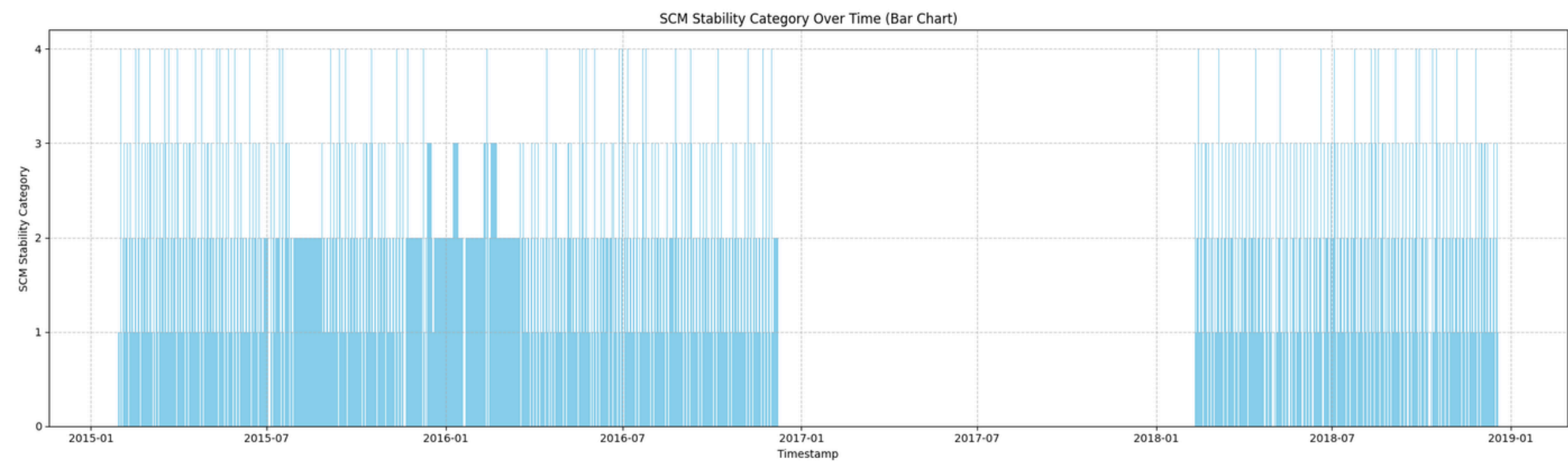
$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

METODOLOGI PENELITIAN

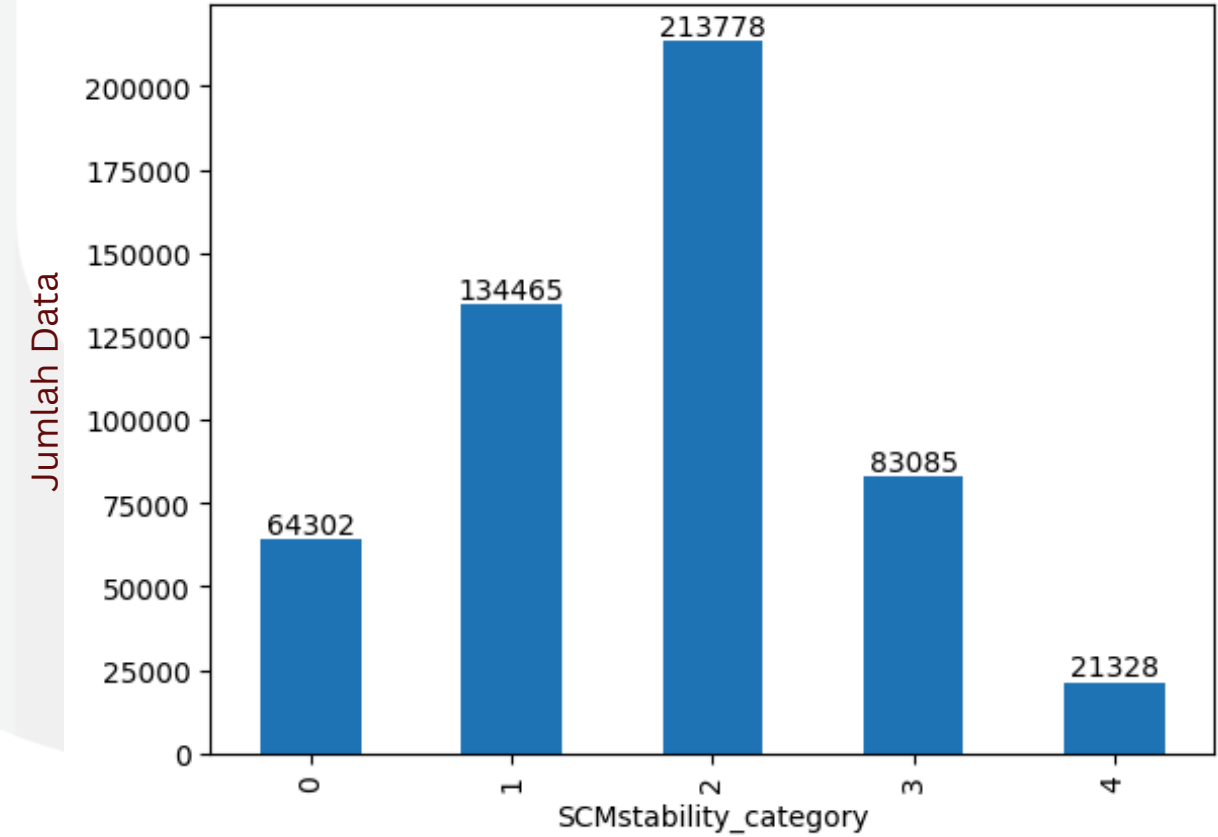


**HASIL
EKSPLORASI
DATA
ANALISIS**

Sebaran Data Berdasarkan Waktu



Sebaran Data Pada Masing-Masing Kelas



HASIL

EKSPLORASI

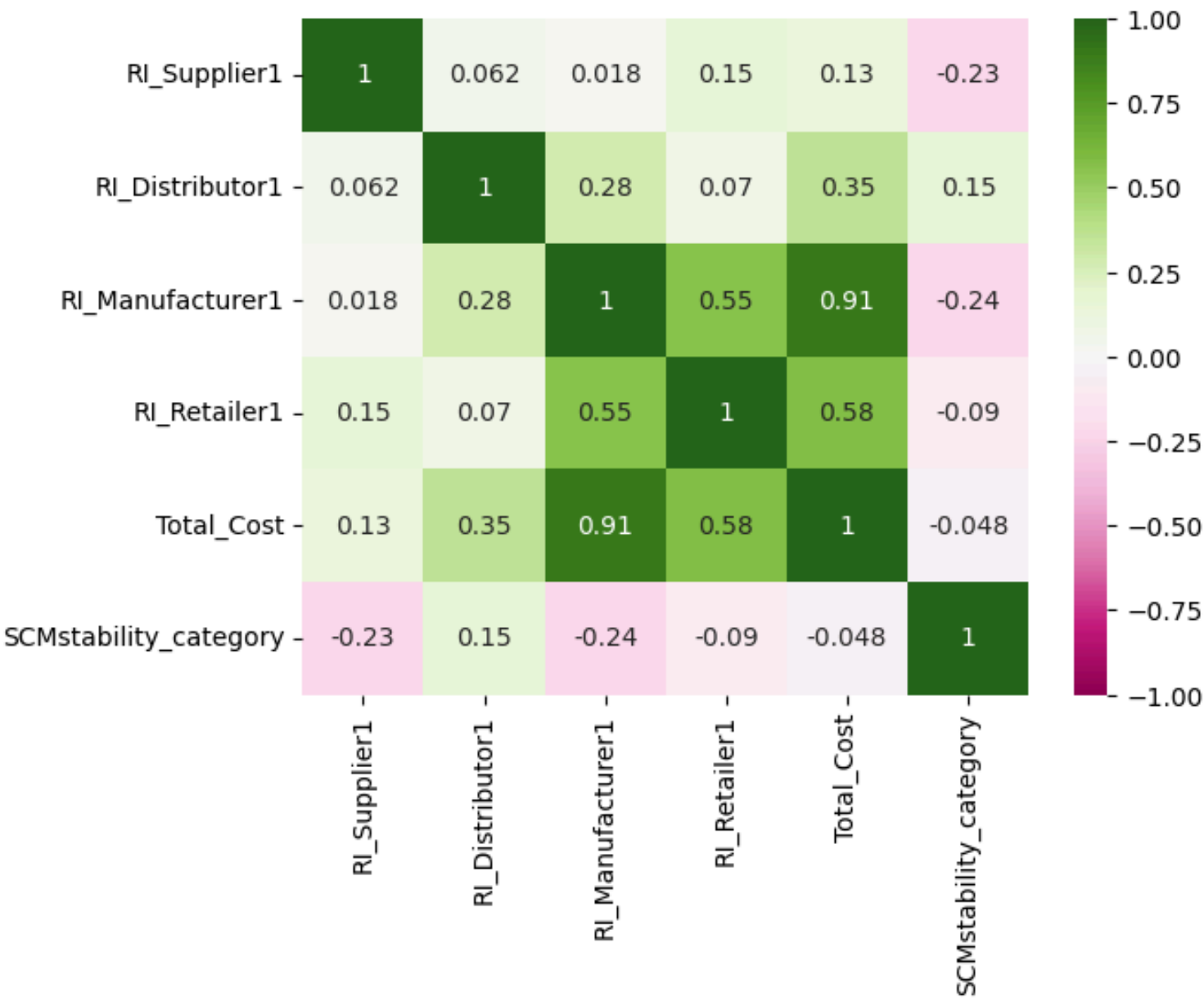
DATA

ANALISIS

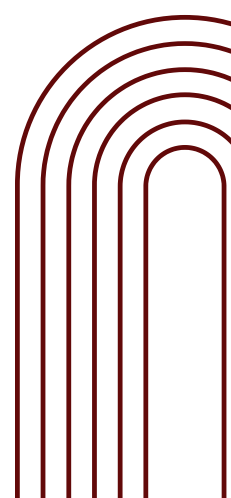
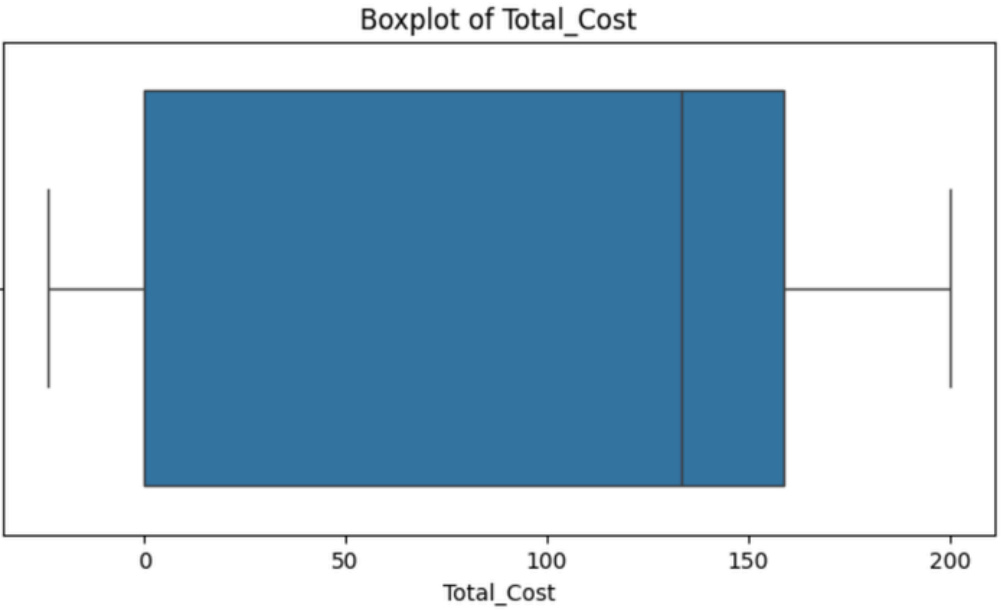
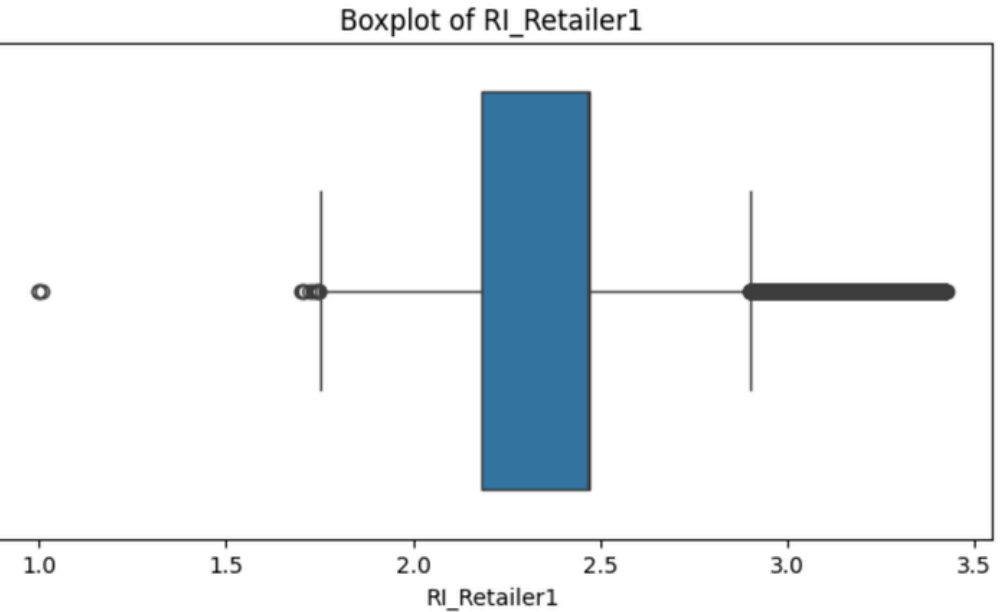
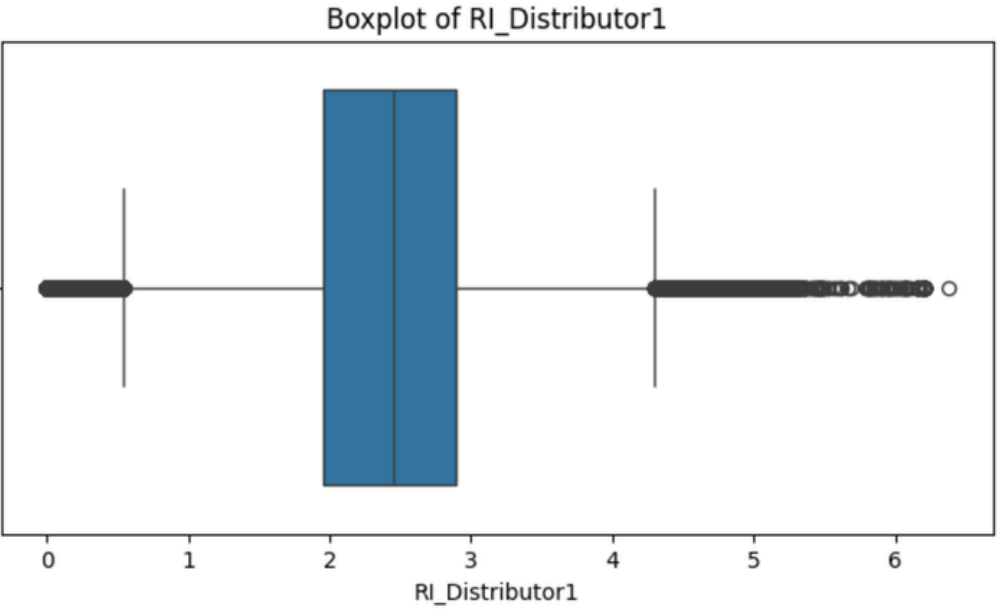
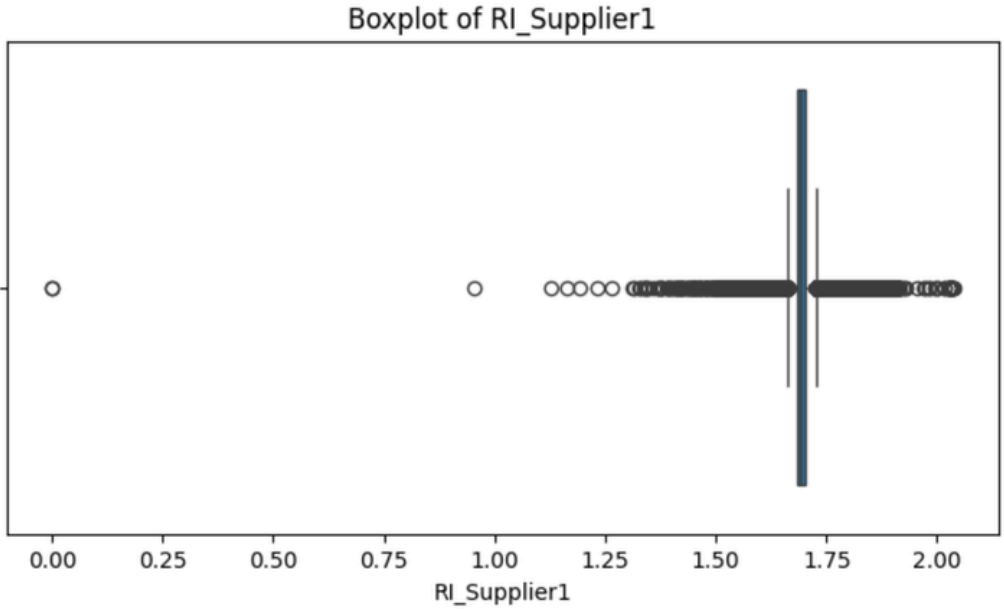
Dataset Heading

	Timestamp	RI_Supplier1	RI_Distributor1	RI_Manufacturer1	RI_Retailer1	Total_Cost	SCMstability_category
0	02/09/2018 12:44:39 PM	1.6396	1.0422	3.1884	3.3609	133.06	4
1	02/09/2018 12:46:39 PM	1.6399	1.0444	3.1877	3.3642	135.02	4
2	02/09/2018 12:48:39 PM	1.6398	1.0422	3.1880	3.3676	135.36	4
3	02/09/2018 12:50:39 PM	1.6403	1.0356	3.1877	3.3705	136.22	4
4	02/09/2018 12:52:39 PM	1.6396	1.0595	3.1888	3.3731	136.48	4

Korelasi Antar Fitur



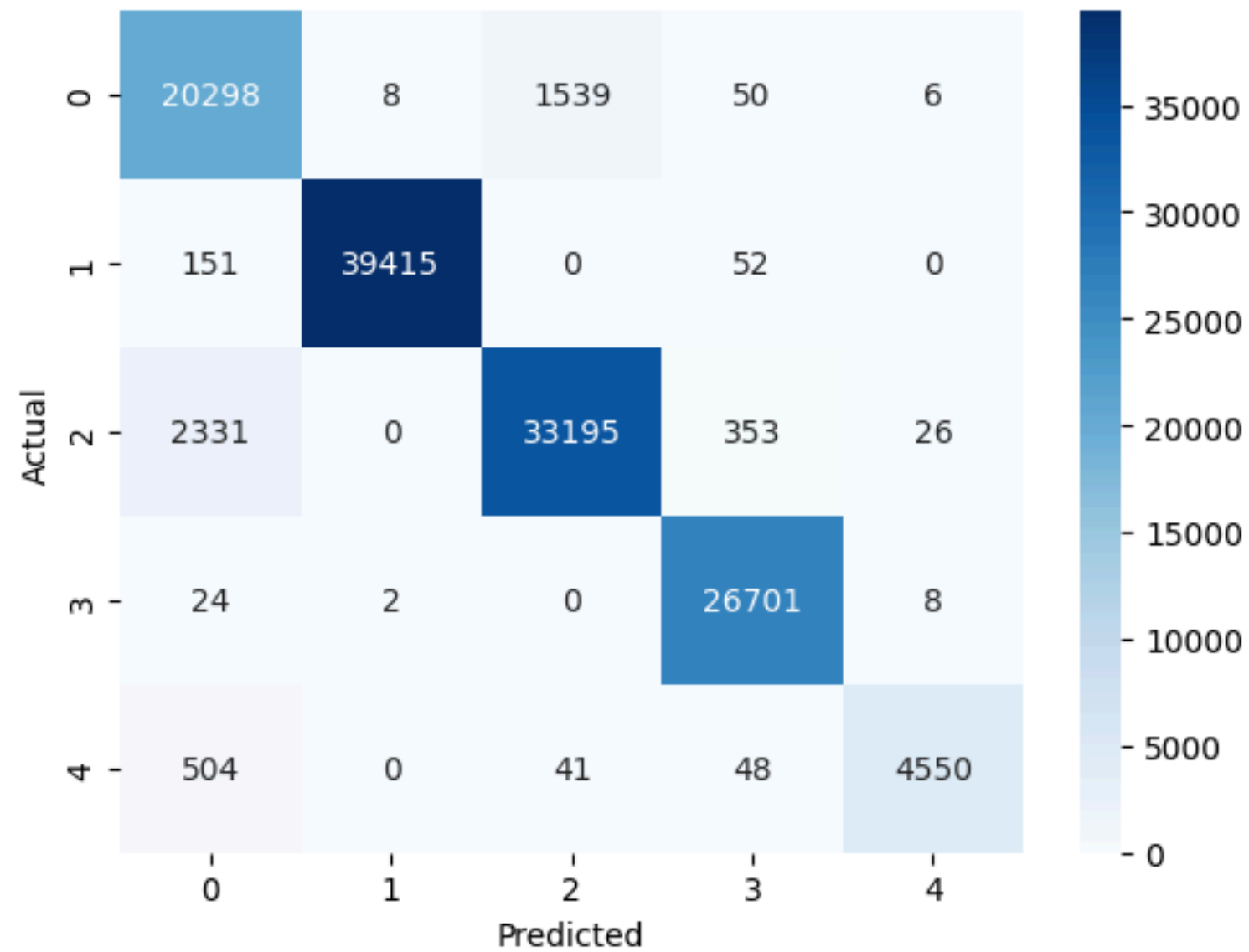
HASIL EKSPLORASI DATA ANALISIS



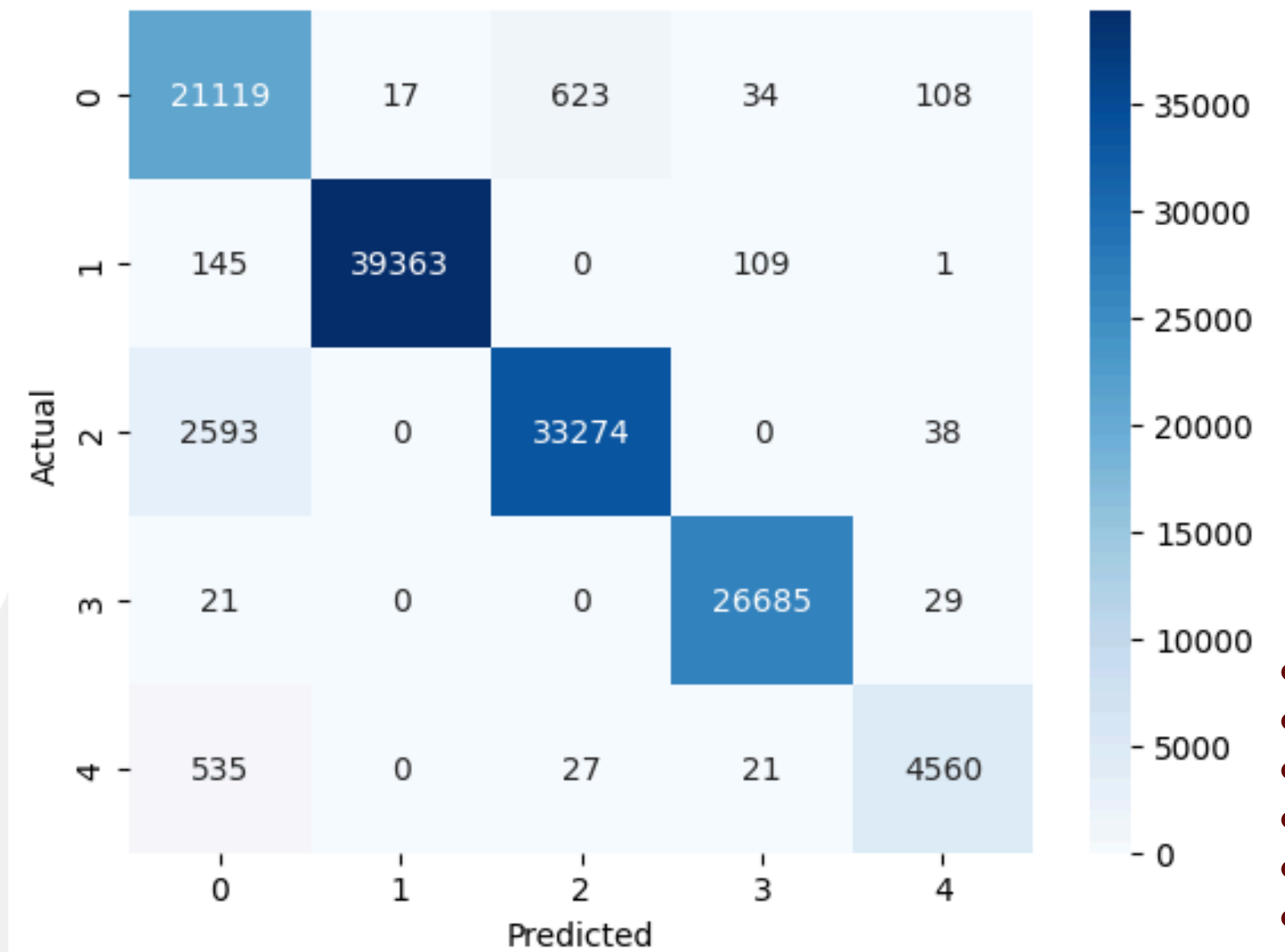
HASIL PENGUJIAN

Normalisasi
Robust vs MinMax

Robust Scaler



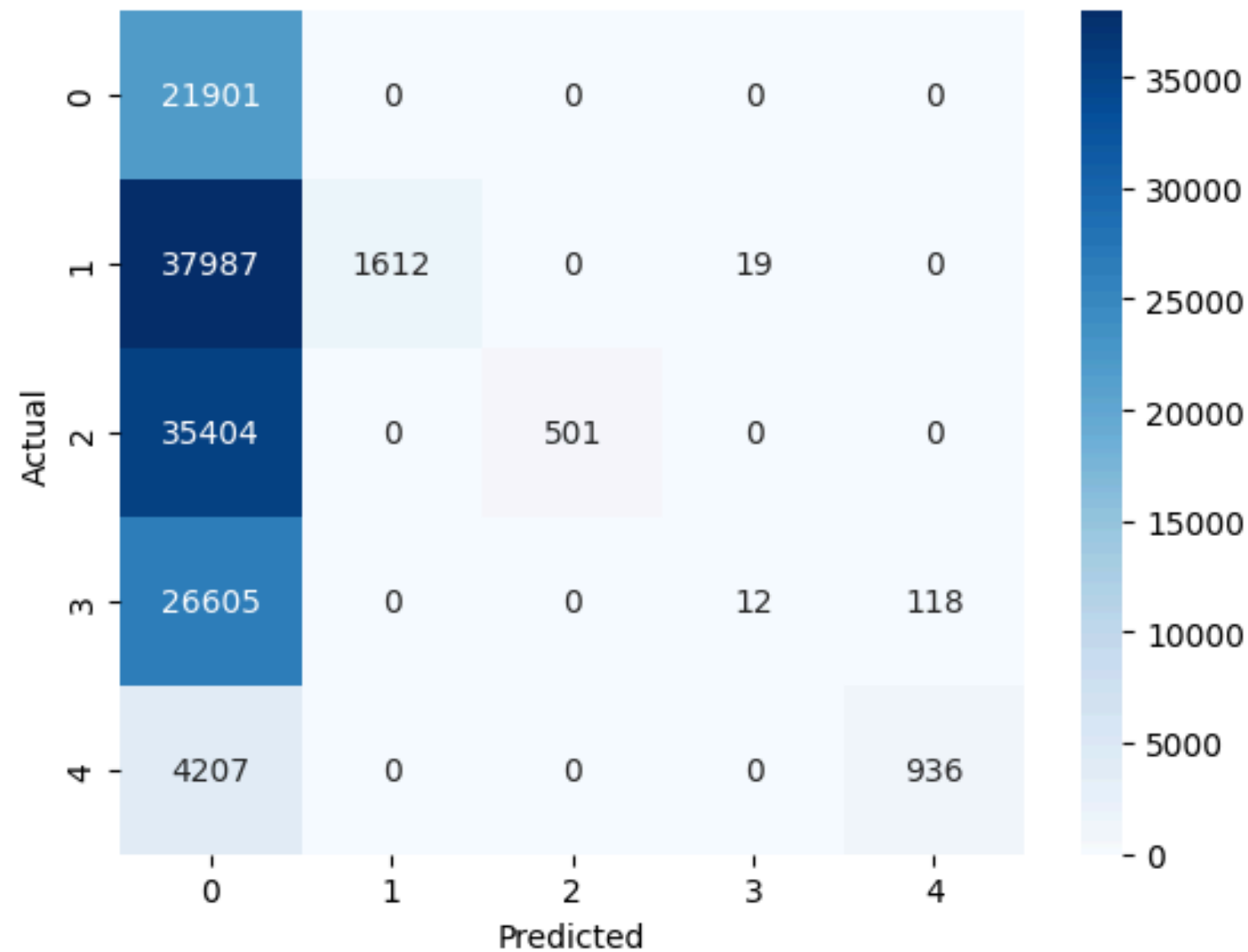
MinMax Scaler



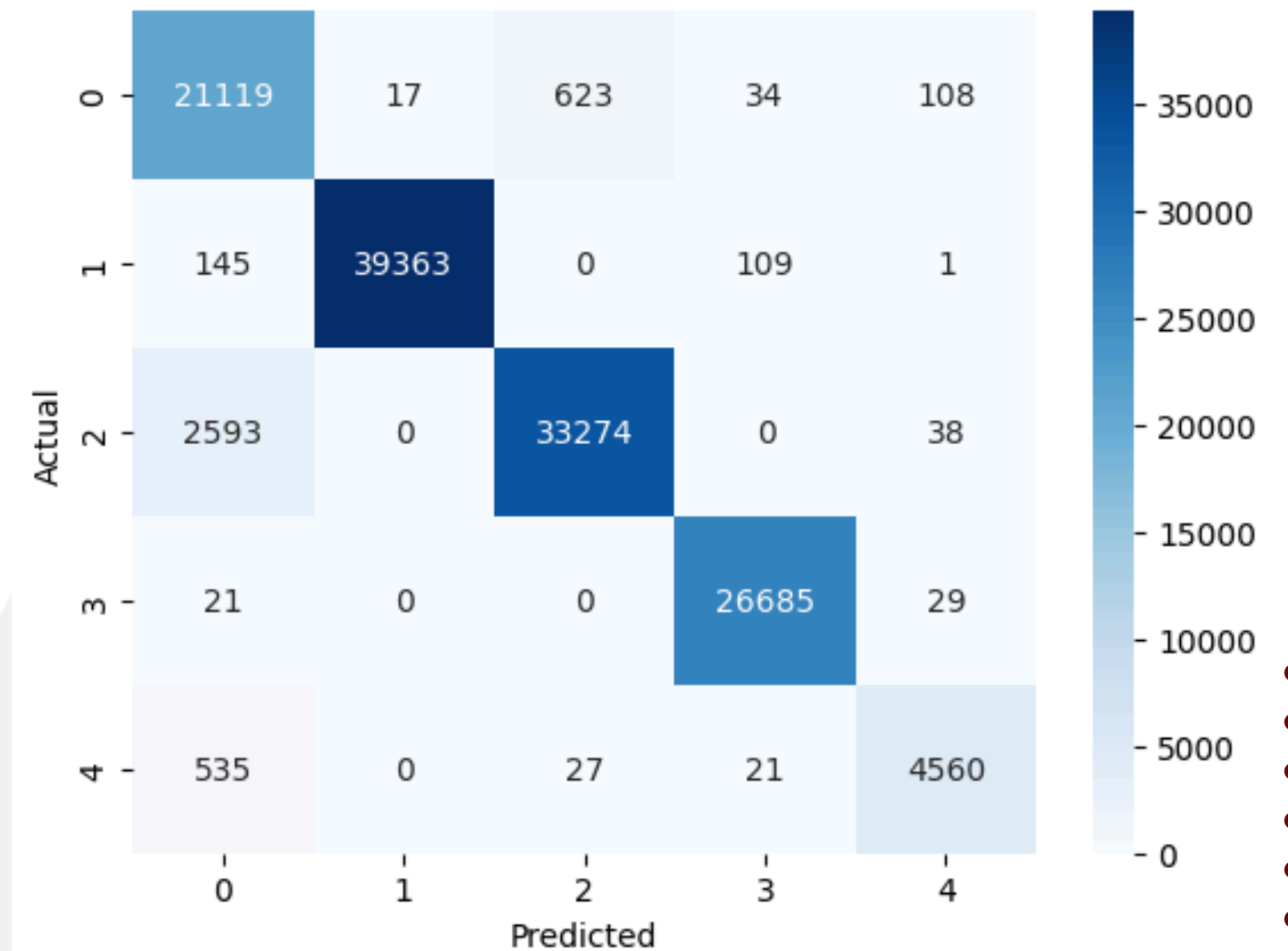
HASIL PENGUJIAN

Balancing vs Tanpa Balancing
Normalisasi MinMax

Dengan Balancing



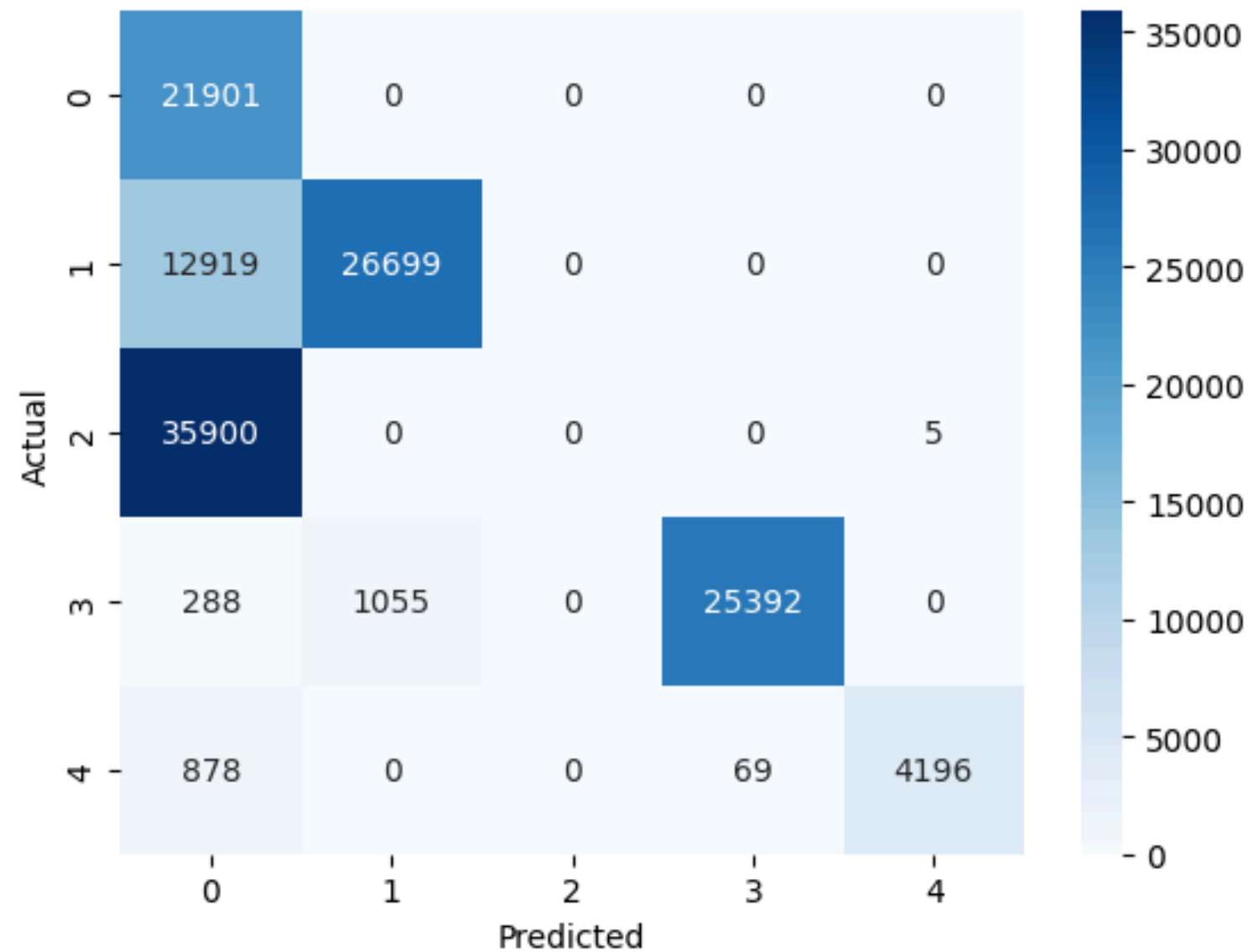
Tanpa Balancing



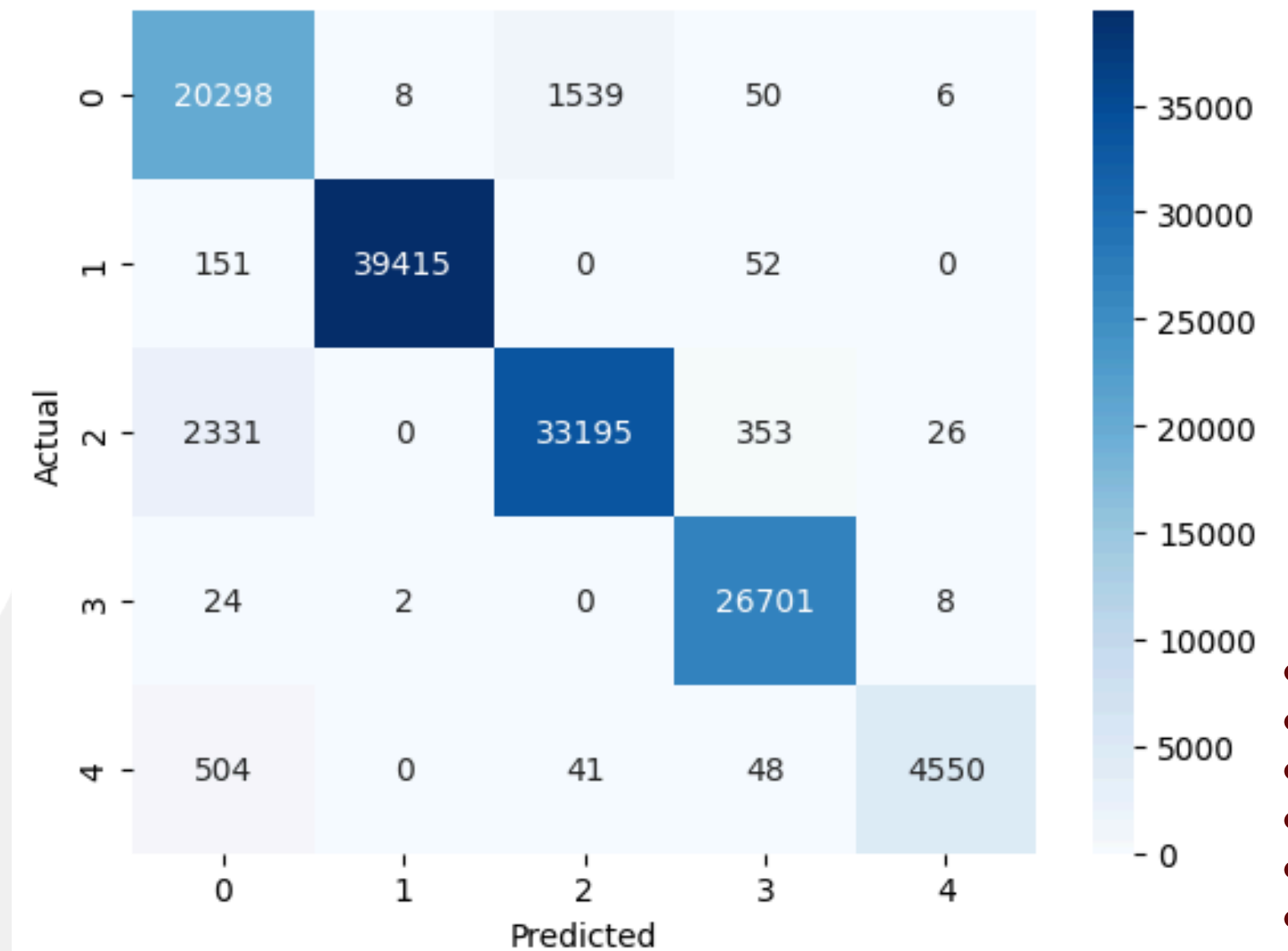
HASIL PENGUJIAN

Balancing vs Tanpa Balancing
Normalisasi Robust

Dengan Balancing

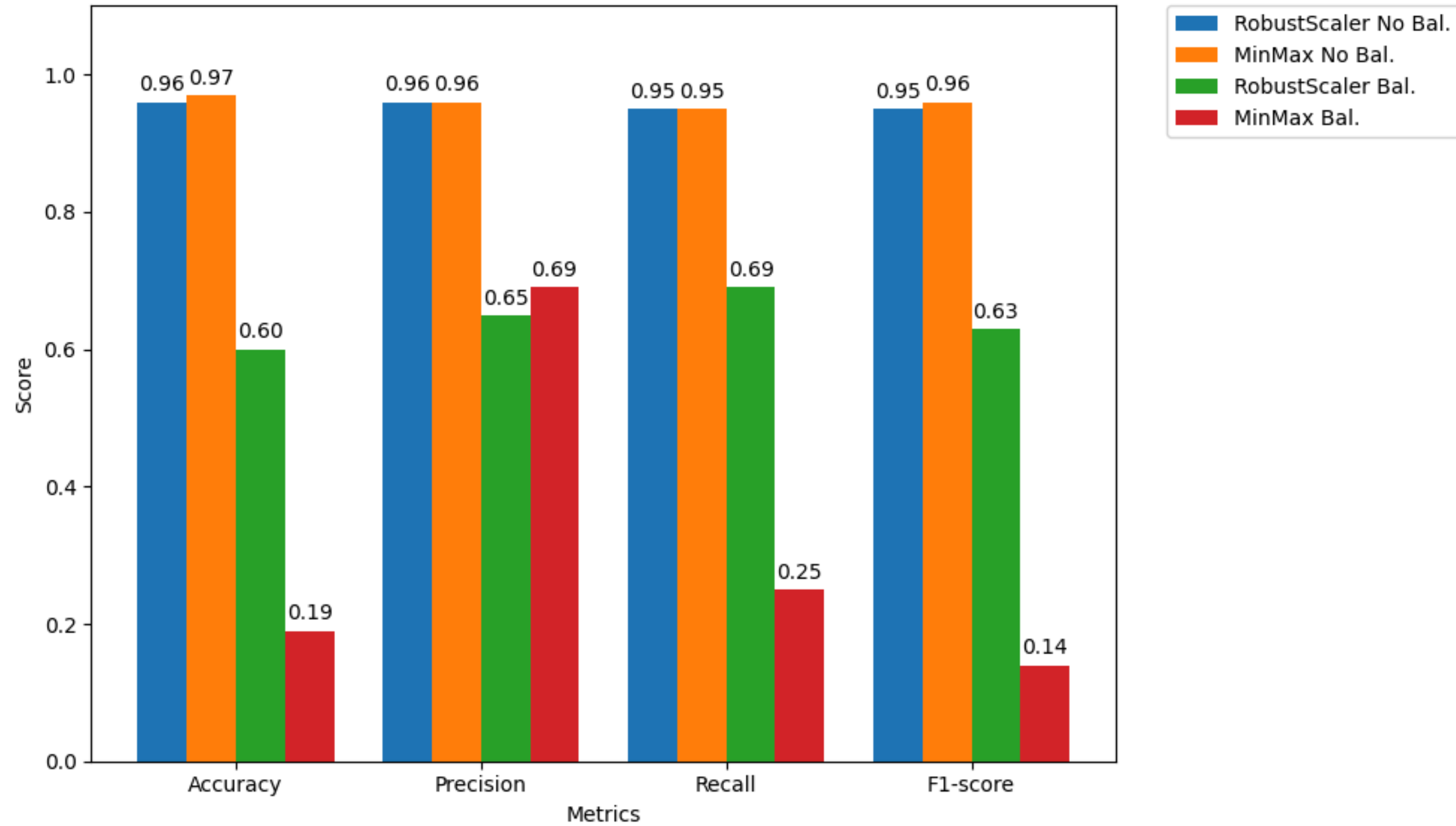


Tanpa Balancing



HASIL PENGUJIAN

Perbandingan Metrik di Berbagai Skenario



KESIMPULAN

1. Data time-series rantai pasok perlu melalui tahapan preprocessing yang tepat, meliputi pembersihan data, normalisasi, dan transformasi ke bentuk tiga dimensi agar sesuai dengan arsitektur LSTM dan dapat dipelajari dengan baik oleh model.
2. Model LSTM mampu memprediksi kategori stabilitas rantai pasok dengan performa terbaik pada skenario normalisasi MinMax Scaler tanpa balancing, sementara perbedaan normalisasi dan penerapan balancing terbukti memengaruhi hasil evaluasi model.



TERIMA KASIH

